Industrial Visit Report: CPVC Pipe and Fittings Manufacturing

Ajay Industrial Corporation Ltd. (Ajay Pipes), Pune | Aarav A. Jain | 2025

1. Purpose of the Visit

The aim of this internship was to understand how a polymer is turned into a pressure-rated engineering product, and how each department in a manufacturing plant contributes to that outcome. I followed the material from the raw-material store through compounding, extrusion, injection moulding, testing, and dispatch, focusing on the engineering decisions made at each stage rather than on the finished products themselves. The visit was especially relevant to my interest in chemical engineering and to my water-treatment projects, where pipe and fitting reliability directly affects whether a system works in the field.

2. Raw-Material Selection and Compounding

Chlorinated polyvinyl chloride (CPVC) is PVC that has undergone further chlorination, which raises its glass transition temperature and allows it to carry hot water without softening. On its own, the resin cannot be processed, so it is blended with heat stabilisers, lubricants, impact modifiers, and pigments. The company sources its CPVC compound from a US-based supplier and uses high-intensity mixing and shearing to achieve uniform dispersion of additives.

I learned that raw-material control starts before anything enters a machine. Incoming batches are checked against specifications, and moisture is a key concern because absorbed water can turn into steam inside the barrel and create voids in the pipe wall. Poorly dispersed stabiliser leaves local regions that degrade faster under heat, which later appear as discolouration or weak spots during testing. Seeing this made it clear that most product failures are decided at the material stage, not at the final inspection.

3. Pipe Extrusion

Pipes are made by twin-screw extrusion. The compound is fed into a heated barrel where two intermeshing screws convey, compress, and melt it, before forcing it through an annular die. CPVC has a narrow processing window: too little heat leaves the material under-fused and brittle, while too much shear or temperature causes thermal degradation and hydrochloric acid release. Twin-screw machines give better control over shear and residence time than single-screw designs, which is why they are preferred for this polymer.

After the die, the hot tube enters a vacuum sizing tank that holds the outside diameter to tolerance while water sprays cool it. A haul-off unit pulls the pipe at a constant speed, and its speed relative to the extruder output sets the wall thickness. The pipe is then printed, cut to length, and, where needed, belled to form a socket. I found it useful to see how one parameter, such as haul-off speed, changes both geometry and residual stress, since faster cooling can lock stress into the wall.

4. Injection Moulding of Fittings and the Tool Room

Fittings such as elbows, tees, and couplers cannot be extruded, so they are injection moulded. Granulated compound is melted and injected under high pressure into steel moulds, held while it cools, and then ejected. The moulds are made from high-strength, corrosion-resistant steel, which is essential because CPVC can release corrosive by-products if overheated. The company uses in-mould grooving to keep socket dimensions consistent so that solvent-cemented joints seat correctly.

The in-house tool room maintains and modifies these moulds. I saw that gate position, cooling-channel layout, and venting all influence weld lines and sink marks, which are common weak points in moulded parts. Fitting design is done in CAD and checked through prototyping before full production, which linked directly to the engineering-design process I had studied in my Leeds coursework.

5. How Pipe Geometry Is Engineered

The most important geometric parameter is the Standard Dimension Ratio (SDR), the outside diameter divided by the minimum wall thickness. A lower SDR means a thicker wall relative to diameter and therefore a higher pressure rating. The governing relationship is the hoop-stress equation:

$$\sigma = P\,(D - e) \,/\, 2e$$

where σ is hoop stress, P is internal pressure, D is outside diameter, and e is wall thickness. Engineers choose a design stress from long-term strength data at the service temperature, apply a safety factor, and then solve for the wall thickness. Because CPVC loses strength as temperature rises, hot-water applications need a thicker wall

than cold-water service at the same pressure. Pressurised supply lines therefore use lower SDRs, while non-pressure drainage lines can use thinner walls relative to diameter and are designed more around stiffness, flow capacity, and resistance to external loads. Socket depth, tolerance on ovality, and chamfer angles are also specified so that joints are as strong as the pipe itself.

6. Quality Control and Testing

Testing was the most extensive part of the visit. Products are tested against ISI/BIS and NSF/ANSI standards, and the company states that its products are tested across 101 parameters. Testing happens at three levels: incoming raw material, in-process checks on the line, and batch testing of finished goods in the laboratory. The main tests I observed or had explained are summarised below.

Test	What it checks	Why it matters
Dimensional check	Outside diameter, wall thickness, ovality, socket depth	Ensures joints fit and the wall carries the rated pressure
Hydrostatic pressure	Pipe held at elevated pressure and temperature for a set duration	Simulates long-term hoop stress; failure means weak resin or thin walls
Vicat softening point	Temperature at which a loaded needle penetrates the sample	Confirms the material can handle hot-water service
Heat reversion	Length change after oven exposure	Reveals frozen-in stress from poor extrusion or cooling
Flattening / crush	Pipe compressed between plates without cracking	Tests ductility and resistance to site handling
Impact resistance	Falling weight dropped on the pipe	Detects brittleness from under-processing or contamination
Opacity and visual	Light transmission, surface finish, voids	Prevents algae growth and flags melt defects

What stood out was that each test targets a specific failure mechanism. Heat reversion, for example, is really a test of the extrusion process, while hydrostatic testing is a test of the material and the geometry together. A batch that fails is traced back through production records to find the cause.

7. Production Planning and Logistics

Production planning schedules which sizes and products run on each line, balancing demand against changeover time, since switching dies or moulds and re-stabilising the process causes downtime and scrap. Finished goods are bundled, labelled for traceability, and moved to warehouses, with the company operating storage hubs in multiple cities including Pune. Proper storage matters technically as well, because pipes stacked unevenly or left in direct sunlight can deform or degrade before use.

8. Reflections and Key Learnings

This visit showed me that manufacturing is a chain of controlled variables, where a small change in moisture, melt temperature, or cooling rate can show up weeks later as a failed pressure test. It also gave me a practical view of polymer chemistry, heat transfer, and stress analysis working together. I later applied this thinking to my filtration and irrigation projects, paying closer attention to material choice and joint design so that the systems could survive real operating conditions rather than just laboratory trials.